

# The Config Bench

Ohm's Law — On-Camera Script | Rev 2

Episode TCB-OHM-001 | John King | @TheConfigBench | Target: <10 min

**Episode Overview:** Builds directly on Lab 3. Opens with a quick schematic review of all three circuits on Qucs (PM1610), then transitions to Xournal++ for the Ohm's Law formula breakdown —  $V=IR$  and all iterations, series resistance, parallel resistance. Values used match exactly what the audience saw on the bench in Labs 1–3. Closes with 5 test questions, oscilloscope giveaway announcement, and a reference to 'Complete Electronics Self-Teaching Guide' Chapter 1 for deeper reading.

**Software Recommendation — Formula Whiteboard:** Use **Xournal++** (free, Ubuntu). Install: `sudo apt install xournalpp`. Run OBS screen capture on PM1610 while you write formulas in real time with mouse or stylus. Gives a natural 'writing it out' feel on camera — far more engaging than a pre-made slide. Switch between Xournal++ and Qucs on the same monitor as needed. Font size 36+ for readability on camera. Use a dark background theme in Xournal++ for contrast.

## Episode Timing Overview

| Segment | Content                                                 | Est. Time | Running Total |
|---------|---------------------------------------------------------|-----------|---------------|
| 1       | Welcome — bridge from Lab 3                             | 0:30      | 0:30          |
| 2       | Qucs review — Circuit 1 (single $47\Omega$ )            | 0:45      | 1:15          |
| 3       | Qucs review — Circuit 2 (two $47\Omega$ series)         | 0:45      | 2:00          |
| 4       | Qucs review — Circuit 3 (4x $470\Omega$ parallel)       | 0:45      | 2:45          |
| 5       | Ohm's Law intro — $V = I \times R$ and all iterations   | 1:30      | 4:15          |
| 6       | Series resistance — $R_t = R_1 + R_2 + \dots$           | 1:00      | 5:15          |
| 7       | Parallel resistance — $R_t = 1/(1/R_1 + 1/R_2 + \dots)$ | 1:15      | 6:30          |
| 8       | Apply formulas back to our three circuits               | 1:00      | 7:30          |
| 9       | Test questions (5) + giveaway + close + book reference  | 1:30      | 9:00          |
|         | Buffer / edit room                                      | 1:00      | ~10:00        |

## Segment 1 — Welcome & Bridge from Lab 3

■ Segment 1 | Est. ~0:30

■ Canon EOS 60D wide overhead. PM1610 glowing in background — Qucs Lab 1 schematic loaded, simulation results visible. Bench clear and clean.

**JOHN:** Welcome back to The Config Bench. I'm John.

**JOHN:** In Lab 3 we ran three circuits, measured voltage and current with two inexpensive meters, and compared those readings to what Qucs calculated. And I told you every time — the numbers will be close but not identical. Tolerances.

**JOHN:** Today we're going to explain exactly why. This is Ohm's Law. And once this clicks, everything on the bench starts making sense.

**JOHN:** We're going to start by pulling up those same three schematics — just as a quick reminder of what we're working with. Then we'll get into the math. Don't worry — it's simpler than it sounds.

**Quick note before we dive in:** This is not a math class — it is an electronics class. But building real circuits means we use math to explain how they behave. I will do my best to walk through each calculation clearly. My one recommendation — have a calculator handy. Your phone calculator works perfectly. We are talking simple arithmetic: multiply, divide, add. That is all Ohm's Law ever asks of you.

## Segment 2 — Qucs Review: Circuit 1 — Single $47\Omega$

■ Segment 2 | Est. ~0:45

■ PM1610 showing Lab 1 Qucs schematic with DC sim results annotated. iPhone close-up on monitor. Pencil ready.

■ QUCS: Lab 1 schematic on screen — battery + single  $47\Omega$  resistor. DC operating point results annotated: ~8.2V node voltage, ~174mA branch current.

**JOHN:** Circuit 1. Single forty-seven ohm resistor. Nine volt battery — measured at eight point two volts.

■ Pencil traces circuit path. Points to voltage annotation, then current annotation on schematic.

**JOHN:** Qucs calculated eight point two volts across the resistor. Current — one hundred and seventy-four milliamps through the circuit. One path, one resistor, maximum current.

**JOHN:** Hold that number. We're coming back to it.

## Segment 3 — Qucs Review: Circuit 2 — Two $47\Omega$ Series

■ Segment 3 | Est. ~0:45

■ Switch PM1610 to Lab 2 Qucs schematic. DC sim results annotated. iPhone close-up on monitor.

■ QUCS: Lab 2 schematic — battery + two  $47\Omega$  resistors in series. Annotated: ~4.1V across each resistor, ~87mA current.

**JOHN:** Circuit 2. Two forty-seven ohm resistors in series. Same battery.

■ Pencil points to voltage split across R1 and R2.

**JOHN:** Qucs shows four point one volts across each resistor. Current dropped to eighty-seven milliamps — exactly half of Circuit 1. We doubled the resistance, current cut in half.

**JOHN:** And the voltage split evenly — four point one plus four point one equals eight point two. Your source voltage. It always has to add up.

## Segment 4 — Qucs Review: Circuit 3 — 4x 470Ω Parallel

■ Segment 4 | Est. ~0:45

■ Switch PM1610 to Lab 3 Qucs schematic. DC sim results annotated. iPhone close-up on monitor.

■ QUCS: Lab 3 schematic — 4x 470Ω in 2S2P configuration. Annotated: ~8.2V across each branch, ~8.7mA per branch, ~17.4mA total supply current.

**JOHN:** Circuit 3. Four four-hundred-and-seventy ohm resistors — two in series per branch, two branches in parallel.

■ Pencil traces both parallel branches. Points to voltage across each, then total current from supply.

**JOHN:** Voltage — eight point two across each branch. Same source voltage, same across every parallel path. Current per branch — eight point seven milliamps. Total from the supply — seventeen point four milliamps. Both branches sharing the load.

**JOHN:** Alright. Three circuits. Three different behaviors. Same battery. Now let's explain why.

## Segment 5 — Ohm's Law: $V = I \times R$ and All Iterations

■ Segment 5 | Est. ~1:30

■ Switch PM1610 from Qucs to Xournal++. Dark background, large font, ready to write. iPhone close-up on monitor. OBS screen capture recording.

■ WHITEBOARD: Open Xournal++ on PM1610. Dark background. Font/pen size large enough to read clearly on camera. Write each formula as you say it — let the audience watch it appear.

**JOHN:** Ohm's Law. One equation. Three variables. Everything on this bench comes back to this.

**JOHN:** And if you have your calculator out — good. I want you to follow along and punch these numbers in as we go. Verify the math yourself. That's how it sticks.

■ Write on Xournal++:  $V = I \times R$ . Large, centered, underlined.

$$V = I \times R$$

**JOHN:** V equals I times R. Voltage equals current times resistance. That's it. Three letters. Everything else is just rearranging them.

**JOHN:** V is voltage — measured in volts. I is current — measured in amps. R is resistance — measured in ohms. Georg Ohm figured this out in 1827. It still runs every circuit on every bench in the world.

■ Write the three iterations beneath the original formula.

$$V = I \times R \quad I = V \div R \quad R = V \div I$$

**JOHN:** Three ways to use the same law. If you know any two values, you can solve for the third.

**JOHN:** Need to find voltage? Multiply current times resistance. Need current? Divide voltage by resistance. Need resistance? Divide voltage by current.

**JOHN:** Here's a trick — the Ohm's Law triangle. Draw a triangle, put V on top, I on the bottom left, R on the bottom right. Cover the one you want to find — what's left tells you the formula. It's on every electrician's toolbelt.

■ Draw a simple V-over-IR triangle in Xournal++.

**JOHN:** We're not going deep on the math today — that's what the textbook is for and we'll get there. Right now I want you to understand what the law is saying. More resistance means less current. Less resistance means more current. Voltage is what pushes it. That's the relationship.

## Segment 6 — Series Resistance: $R_t = R_1 + R_2 + R_3...$

■ Segment 6 | Est. ~1:00

■ Still on Xournal++. Clear screen or new page. Write series formula as you speak.

**JOHN:** Series circuits first. When resistors are in series — one after another in a single path — you add them up. Total resistance is just the sum.

$$R_t = R_1 + R_2 + R_3 \dots$$

■ Write Circuit 2 values beneath:  $R_t = 47\Omega + 47\Omega = 94\Omega$

**JOHN:** In Circuit 2 — two forty-seven ohm resistors in series. Forty-seven plus forty-seven equals ninety-four ohms total.

**JOHN:** Now apply Ohm's Law. I equals V divided by R. Eight point two volts divided by ninety-four ohms.

■ Write:  $I = 8.2V \div 94\Omega = 0.087A = 87mA$

$$I = 8.2V \div 94\Omega = 0.087A = 87mA$$

**JOHN:** Eighty-seven milliamps. That is exactly what Qucs showed us. And it's very close to what the ANENG measured on the bench. The law works.

**JOHN:** In series — current is the same everywhere in the loop. Voltage divides across each resistor proportional to its value. Equal resistors — equal voltage split.

## Segment 7 — Parallel Resistance: $R_t = 1 \div (1/R_1 + 1/R_2 + \dots)$

■ Segment 7 | Est. ~1:15

■ New page in Journal++. Write parallel formula as you explain it. Take it slowly — this is the one that trips beginners.

**JOHN:** Parallel circuits are a little different. When resistors are in parallel — multiple paths for current to flow — the total resistance goes DOWN. More paths, less overall resistance.

**JOHN:** The formula looks more complicated but the concept is simple.

$$R_t = 1 \div (1/R_1 + 1/R_2 + 1/R_3 \dots)$$

**JOHN:** The total resistance equals one divided by — the sum of one-over-each-resistor. It's called the reciprocal method. Don't let it intimidate you.

■ Write Circuit 3 values. Each branch =  $470\Omega + 470\Omega = 940\Omega$ . Two branches in parallel.

**JOHN:** In Circuit 3 — each branch has two four-seventy ohm resistors in series. So each branch is nine hundred and forty ohms. Two branches in parallel.

■ Write:  $R_t = 1 \div (1/940 + 1/940) = 1 \div (0.00213) = 470\Omega$

$$R_t = 1 \div (1/940 + 1/940) = 1 \div 0.00213 = 470\Omega$$

**JOHN:** Total resistance — four hundred and seventy ohms. Now apply Ohm's Law.

■ Write:  $I = 8.2V \div 470\Omega = 0.0174A = 17.4mA$

$$I = 8.2V \div 470\Omega = 0.0174A = 17.4mA$$

**JOHN:** Seventeen point four milliamps total. Exactly what Qucs showed. Split evenly between two branches — eight point seven milliamps each.

**JOHN:** Here's the key takeaway for parallel: voltage stays the same across every branch. Current splits. And total resistance is always less than the smallest individual branch.

## Segment 8 — Tying It All Together

■ Segment 8 | Est. ~1:00

■ Switch PM1610 back to Qucs — cycle through all three schematics with sim results visible. Quick cuts. Pencil points to key values.

**JOHN:** Now go back to those three circuits and it all makes sense.

■ PM1610 — Circuit 1. Pencil points to current annotation.

**JOHN:** Circuit 1 — forty-seven ohms total. Eight point two volts. Current — one seventy-four milliamps.  $V$  equals  $I$  times  $R$ . Eight point two equals zero point one seven four times forty-seven. Check.

■ PM1610 — Circuit 2. Pencil points to series values.

**JOHN:** Circuit 2 — ninety-four ohms. Half the current. Voltage split evenly across two equal resistors. Ohm's Law. Check.

■ PM1610 — Circuit 3. Pencil points to parallel branches and total current.

**JOHN:** Circuit 3 — four hundred and seventy ohms total. Seventeen point four milliamps from the supply. Voltage identical across both branches. Ohm's Law. Check.

**JOHN:** And those slight differences between what Qucs calculated and what your meters showed on the bench? That's component tolerances and meter tolerances. The law is still holding. The math still works. Your bench is just real life.

## Segment 9 — Test Questions, Giveaway & Close

■ Segment 9 | Est. ~1:30

■ Canon wide. PM1610 showing Xournal++ with  $V=IR$  triangle visible. Relaxed, direct to camera.

**JOHN:** Alright — five questions. Based on the exact circuits and values we've been using. Drop your answers in the comments. I'll announce the winner of the DSO152 oscilloscope from the Lab 3 entries very soon — so if you haven't commented yet, go do it now.

### Test Questions — Based on Lab 1–3 Circuit Values

**Q1:** A 9V battery (measured at 8.2V) is connected across a single  $47\Omega$  resistor. Using Ohm's Law, what is the current through the circuit?

- A) 87 mA
- B) 174 mA
- C) 17.4 mA
- D) 47 mA

■ Answer: B) 174 mA ( $I = V \div R = 8.2 \div 47 = 0.1745A \approx 174mA$ )

**Q2: In Circuit 2, two  $47\Omega$  resistors are connected in series. What is the total resistance?**

- A)  $47\Omega$
- B)  $23.5\Omega$
- C)  $94\Omega$
- D)  $188\Omega$

■ **Answer: C)  $94\Omega$  ( $R_t = R_1 + R_2 = 47 + 47 = 94\Omega$ )**

**Q3: In Circuit 2 with  $R_t = 94\Omega$  and  $V = 8.2V$ , how much voltage appears across each resistor?**

- A)  $8.2V$  across each
- B)  $2.05V$  across each
- C)  $4.1V$  across each
- D)  $1.0V$  across each

■ **Answer: C)  $4.1V$  across each (Equal resistors in series split voltage equally —  $8.2 \div 2 = 4.1V$ )**

**Q4: In Circuit 3, each parallel branch has two  $470\Omega$  resistors in series. What is the resistance of each branch?**

- A)  $235\Omega$
- B)  $470\Omega$
- C)  $940\Omega$
- D)  $1880\Omega$

■ **Answer: C)  $940\Omega$  ( $R_t$  per branch =  $470 + 470 = 940\Omega$ )**

**Q5: In a parallel circuit, if you add a second identical branch alongside the first, what happens to the total resistance?**

- A) It doubles
- B) It stays the same
- C) It is cut in half
- D) It increases by the resistance of the new branch

■ **Answer: C) It is cut in half (Two identical parallel branches:  $R_t = R/2$ . More paths = less total resistance.)**

## On-Camera Close

■ *Canon wide. Direct to camera. Relaxed close.*

**JOHN:** That's Ohm's Law.  $V$  equals  $I$  times  $R$ . Three variables, one relationship, infinite applications. Everything you build on this bench runs on it.

**JOHN:** If you want to go deeper — and I strongly recommend you do — pick up a copy of 'Complete Electronics Self-Teaching Guide with Projects' by Boysen and Kybett. Chapter 1 goes into Ohm's Law in much greater detail with worked problems and exercises. It's the textbook I use here on the bench. Link in the description.

**JOHN:** Drop your answers to the five questions in the comments. The winner of the FNIRSI DSO152 oscilloscope from Lab 3 will be announced very soon — if you haven't entered yet, go back and leave a comment on that video.

**JOHN:** Subscribe so you don't miss the next lab. I'm John. This is The Config Bench. Thanks for watching.

## On-Camera Reference — Ohm's Law Values

| Circuit | Config         | Rt   | V (measured)       | I (calculated) | I per branch |
|---------|----------------|------|--------------------|----------------|--------------|
| 1       | Single 47Ω     | 47Ω  | 8.2V               | 174 mA         | —            |
| 2       | Two 47Ω series | 94Ω  | 8.2V (4.1V each R) | 87 mA          | —            |
| 3       | 4x 470Ω (2S2P) | 470Ω | 8.2V each branch   | 17.4 mA total  | 8.7 mA each  |

## YouTube Metadata

| Field           | Content                                                                                                                                                                                                                                                                              |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Title Option 1  | Ohm's Law Explained — V=IR on a Real Bench   The Config Bench                                                                                                                                                                                                                        |
| Title Option 2  | Why Your Meter Doesn't Match the Simulation — Ohm's Law   TCB                                                                                                                                                                                                                        |
| Title Option 3  | Ohm's Law: Series, Parallel & the Math Behind Your Bench   TCB                                                                                                                                                                                                                       |
| Description     | We break down Ohm's Law using the exact circuits from Labs 1-3. V=IR, series resistance, parallel resistance — explained with Qucs schematics and real bench values. Includes 5 test questions. Reference: 'Complete Electronics Self-Teaching Guide' by Boysen & Kybett, Chapter 1. |
| Tags            | Ohm's Law, V=IR, series circuit, parallel circuit, electronics for beginners, Qucs simulator, TheConfigBench, breadboard, 47 ohm, 470 ohm, electronics math, resistor                                                                                                                |
| End Screen      | Link to Lab 3, Lab 1, Subscribe button                                                                                                                                                                                                                                               |
| Pinned Comment  | 5 Test Questions are in the video — drop your answers below! DSO152 giveaway winner announced very soon from Lab 3 comments.                                                                                                                                                         |
| Chapter Markers | 0:00 Intro   0:30 Circuit 1 Review   1:15 Circuit 2 Review   2:00 Circuit 3 Review   2:45 Ohm's Law V=IR   4:15 Series Resistance   5:15 Parallel Resistance   6:30 Putting It Together   7:30 Test Questions & Close                                                                |
| Book Reference  | 'Complete Electronics Self-Teaching Guide with Projects' — Boysen & Kybett, ISBN 1118217322, Chapter 1                                                                                                                                                                               |